

# INTRODUCTION

## CAD/CAM

CAD/CAM is a term which means computer-aided design and computer-aided manufacturing. It is the technology concerned with the use of digital computers to perform certain functions in design and production. This technology is moving in the direction of greater integration of design and manufacturing, two activities which have traditionally been treated as distinct and separate functions in a production firm. Ultimately, CAD/CAM will provide the technology base for the computer-integrated factory of the future.

Computer-aided design (CAD) can be defined as the use of computer systems to assist in the creation, modification, analysis, or optimization of a design. The computer systems consist of the hardware and software to perform the specialized design functions required by the particular user firm. The CAD hardware typically includes the computer, one or more graphics display terminals, keyboards, and other peripheral equipment. The CAD software consists of the computer programs to implement computer graphics on the system plus application programs to facilitate the engineering functions of the user company. Examples of these application programs include stress-strain analysis of components, dynamic response of mechanisms, heat-transfer calculations, and numerical control part programming. The collection of application programs will vary from one user firm to the next because their product lines, manufacturing processes, and customer markets are different. These factors give rise to differences in CAD system requirements.

Computer-aided manufacturing (CAM) can be defined as the use of computer systems to plan, manage, and control the operations of a manufacturing plant through either direct or indirect computer interface with the plant's production resources. As indicated by the definition, the applications of computer-aided manufacturing fall into two broad categories:

1. *Computer monitoring and control.* These are the direct applications in which the computer is connected directly to the manufacturing process for the purpose of monitoring or controlling the process.

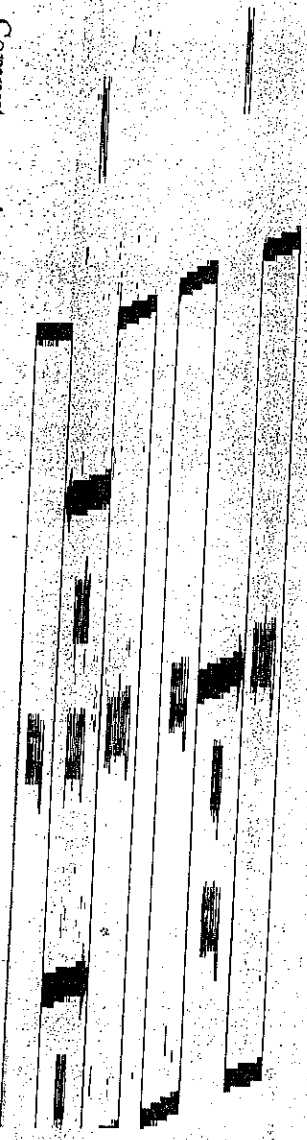
2. *Manufacturing support applications.* These are the indirect applications in which the computer is used in support of the production operations in the plant, but there is no direct interface between the computer and the manufacturing process.

The distinction between the two categories is fundamental to an understanding of computer-aided manufacturing. It seems appropriate to elaborate on our brief definitions of the two types.

Computer monitoring and control can be separated into monitoring applications and control applications. Computer process monitoring involves a direct computer interface with the manufacturing process for the purpose of observing the process and associated equipment and collecting data from the process. The computer is not used to control the operation directly. The control of the process remains in the hands of human operators, who may be guided by the information compiled by the computer.

Computer process control goes one step further than monitoring by not only observing the process but also controlling it based on the observations. The distinction between monitoring and control is displayed in Figure. With computer monitoring the flow of data between the process and the computer is in one direction only, from the process to the computer. In control, the computer interface allows for a two-way flow of data. Signals are transmitted from the process to the computer, just as in the case of computer monitoring. In addition, the computer issues command signals directly to the manufacturing process based on control algorithms contained in its software.

In addition to the applications involving a direct computer-process interface for the purpose of process monitoring and control, computer-aided manufacturing also includes indirect applications in which the computer serves a support role in the manufacturing operations of the plant. In these applications, the computer is not linked directly to the manufacturing process.



Computer monitoring versus computer control: (a) computer monitoring;

(b) computer control.

Instead, the computer is used "off-line" to provide plans, schedules, forecasts, instructions, and information by which the firm's production resources can be managed more effectively. The form of the relationship between the computer and the process is represented symbolically in Figure. Dashed lines are used to indicate that the communication and control link is an off-line connection, with human beings often required to consummate the interface. Some examples of CAM for manufacturing support that are discussed in subsequent chapters of this book include:

*Numerical control part programming by computers.* Control programs are prepared for automated machine tools.

*Computer-automated process planning.* The computer prepares a listing of the operation sequence required to process a particular product or component.

*Computer-generated work standards.* The computer determines the time standard for a particular production operation.

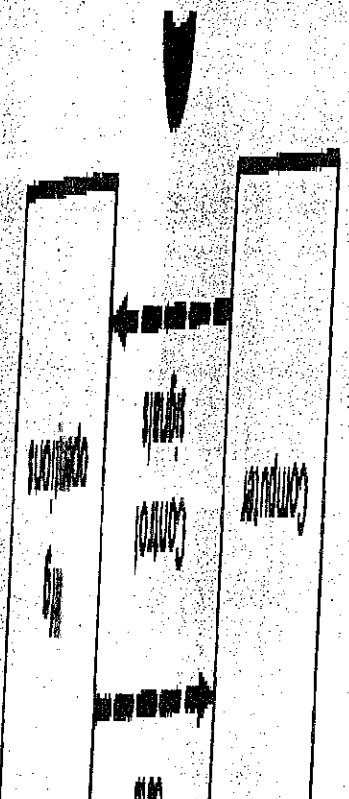
*Production scheduling.* The computer determines an appropriate schedule for meeting production requirements.

*Material requirements planning.* The computer is used to determine when to order raw materials and purchased components and how many should be ordered to achieve the production schedule.

*Shop floor control.* In this CAM application, data are collected from the factory to determine progress of the various production shop orders.

In all of these examples, human beings are presently required in the

application either to provide input to the computer programs or to interpret the computer output and implement the required action.



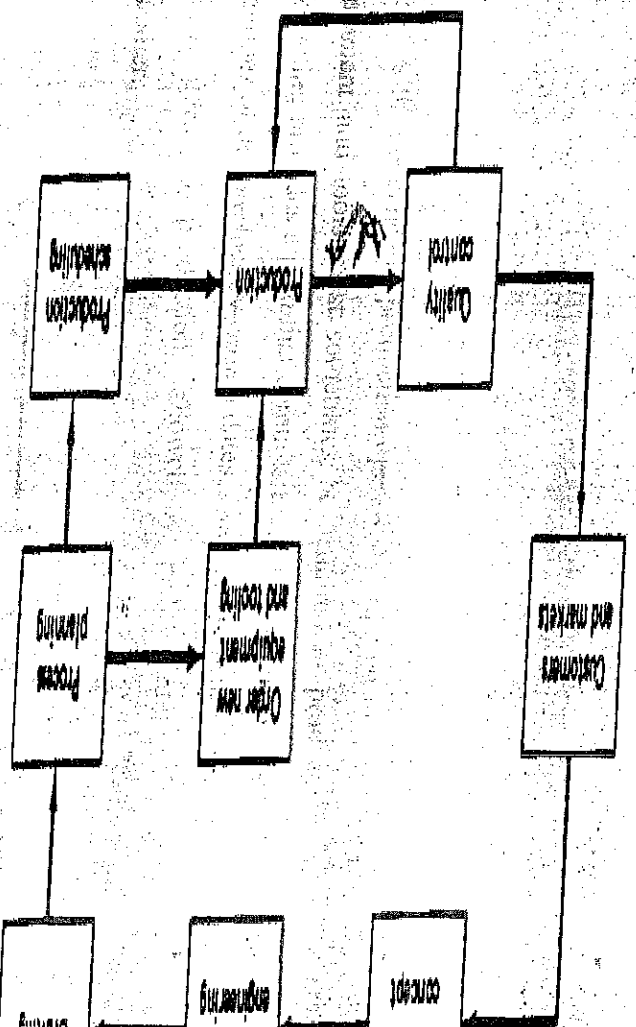
## THE PRODUCT CYCLE AND CAD/CAM

For the reader to appreciate the scope of CAD/CAM in the operations of a manufacturing firm, it is appropriate to examine the various activities and functions that must be accomplished in the design and manufacture of a product. We will refer to these activities and functions as the product cycle.

A diagram showing the various steps in the product cycle is presented in Figure. The cycle is driven by customers and markets which demand the product. It is realistic to think of these as a large collection of diverse industrial and consumer markets rather than one monolithic market. Depending on the particular customer group, there will be differences in the way the product cycle is activated. In some cases, the design functions are performed by the customer and the product is manufactured by a different firm. In other cases, design and manufacturing is accomplished by the same firm. Whatever the case, the product cycle begins with a concept, an idea for a product. This concept is cultivated, refined, analyzed, improved, and translated into a plan for the product through the design engineering process. The plan is documented by drafting. A set of engineering drawings showing how the product is made and providing a set of specifications indicating how the product should perform.

Except for engineering changes which typically follow the product throughout its life cycle, this completes the design activities in Figure. The next activities involve the manufacture of the product. A process plan is formulated which

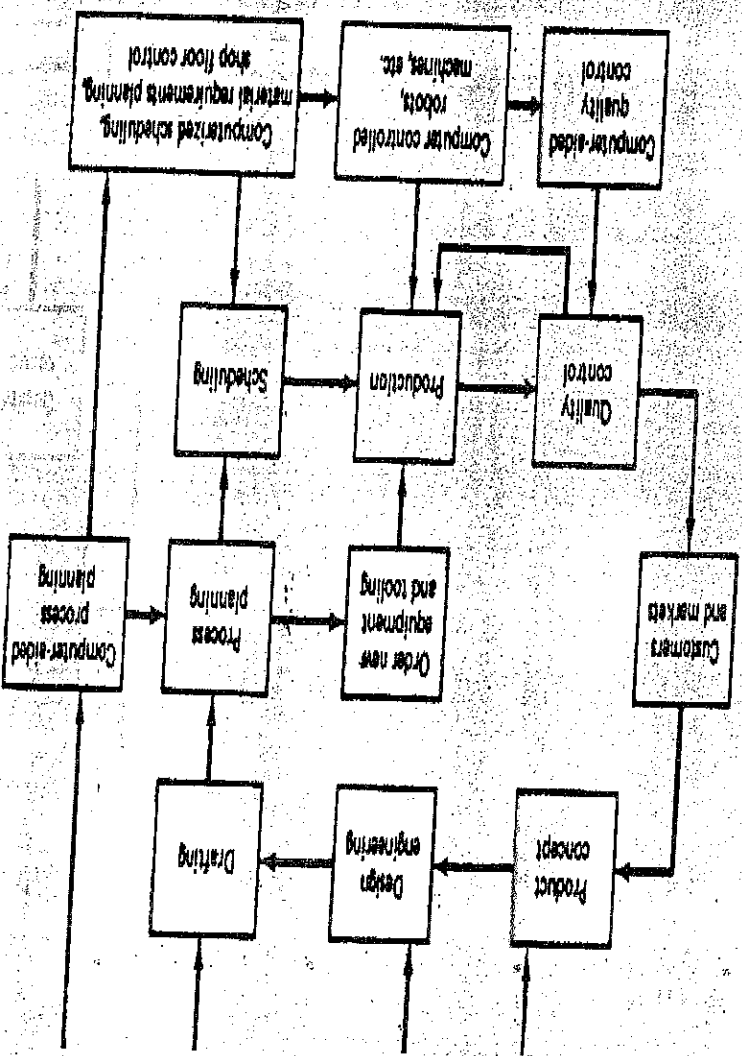
specifies the sequence of production operations required to make the product. New equipment and tools must sometimes be acquired to produce the new product. Scheduling provides a plan that commits the company to the manufacture of certain quantities of the product by certain dates. Once all of these plans are formulated, the product goes into production, followed by quality testing, and delivery to the customer.



Product cycle (design and manufacturing).

The impact of CAD/CAM is manifest in all of the different activities in the product cycle, as indicated in Figure. Computer-aided design and automated drafting are utilized in the conceptualization, design, and documentation of the product. Computers are used in process planning and scheduling to perform these functions more efficiently. Computers are used in production to monitor and control the manufacturing operations. In quality control, computers are used to perform inspections and performance tests on the product and its components.

As illustrated in Figure, CAD/CAM is overlaid on virtually all of the activities and functions of the product cycle. In the design and production operations of a modern manufacturing firm, the computer has become a pervasive, useful, and indispensable tool. It is strategically important and competitively imperative that manufacturing firms and the people who are employed by them understand CAD/



Product cycle revised with CAD/CAM overlaid.

## AUTOMATION AND CAD/CAM

Automation is defined as the technology concerned with the application of complex mechanical, electronic, and computer-based systems in the operation and control of production. It is the purpose of this section to establish the relationship between CAD/CAM and automation.

As indicated in previous Section, there are differences in the way the product cycle is implemented for different firms involved in production. Production activity can be divided into four main categories:

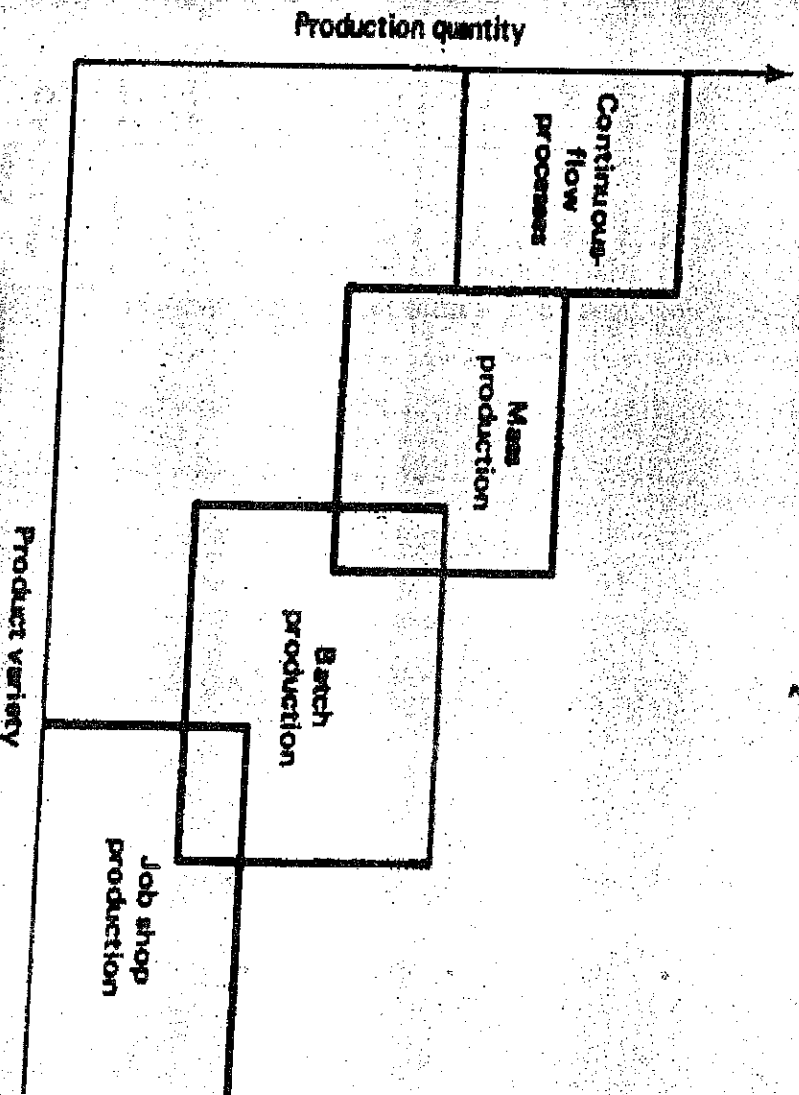
1. Continuous-flow processes
2. Mass production of discrete products
3. Batch production
4. Job shop production

The definitions of the four types are given in Table. The relationships among the four types in terms of product variety and production quantities can be conceptualized as shown in Figure. There is some overlapping of the categories as the figure indicates. Table provides a list of some of the notable achievements in automation technology for each of the four production types.

One fact that stands out from Table is the importance of computer technology in automation. Most of the automated production systems implemented today make use of computers. This connection between the digital computer and manufacturing automation may seem perfectly logical to the reader. However, this logical connection has not always existed. For one thing, automation technology

**TABLE Four Types of Production**

| Category                                | Description                                                                                                                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Continuous-flow processes            | Continuous dedicated production of large amounts of bulk product. Examples include continuous chemical plants and oil refineries                                                                                      |
| 2. Mass production of discrete products | Dedicated production of large quantities of one product (with perhaps limited model variations). Examples include automobiles, appliances, and engine blocks.                                                         |
| 3. Batch production                     | Production of medium lot sizes of the same product or component. The lots may be produced once or repeated periodically. Examples include books, clothing, and certain industrial machinery.                          |
| 4. Job shop production                  | Production of low quantities, often one of a kind, of specialized products. The products are often customized and technologically complex. Examples include prototypes, aircraft, machine tools, and other equipment. |



Four production types related to quantity and product variation

**TABLE Automation Achievements for the Four Types of Production**

| Category                                | Automation achievements                                                                                                                                                                                                    |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Continuous-flow processes            | <p>Flow process from beginning to end</p> <p>Sensor technology available to measure important process variables</p> <p>Use of sophisticated control and optimization strategies</p> <p>Fully computer-automated plants</p> |
| 2. Mass production of discrete products | <p>Automated transfer machines</p> <p>Dial indexing machines</p> <p>Partially and fully automated assembly lines</p> <p>Industrial robots for spot welding, parts handling, machine loading, spray painting, etc.</p>      |

|                        |                                                                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | Automated materials handling systems<br>Computer production monitoring                                                                                                                                                 |
| 3. Batch production    | Numerical control (NC), direct numerical control (DNC),<br>computer numerical control (CNC)<br>Adaptive control machining<br>Robots for arc welding, parts handling, etc.<br>Computer-integrated manufacturing systems |
| 4. Job shop production | Numerical control, computer numerical control                                                                                                                                                                          |

## FUNDAMENTALS OF CAD

### INTRODUCTION

The computer has grown to become essential in the operations of business, government, the military, engineering, and research. It has also demonstrated itself, especially in recent years, to be a very powerful tool in design and manufacturing. In this and the following two chapters, we consider the application of computer technology to the design of a product. This section provides an overview of computer-aided design.

#### **The CAD system defined**

As defined in previous section, computer-aided design involves any type of design activity which makes use of the computer to develop, analyze, or modify an engineering design. Modern CAD systems (also often called CAD/CAM systems) are based on interactive computer graphics (ICG). Interactive computer graphics denotes a user-oriented system in which the computer is employed to create, transform, and display data in the form of pictures or symbols. The user in the computer graphics design system is the designer, who communicates data and commands to the computer through any of several input devices. The computer communicates with the user via a cathode ray tube (CRT). The designer creates an image on the CRT screen by entering commands to call the desired software sub-routines stored in the computer. In most systems, the image is constructed out of basic geometric elements- points, lines, circles, and so on. It can be modified according to the commands of the designer- enlarged, reduced in size, moved to another location on the screen, rotated,

and other transformations. Through these various manipulations, the required details of the image are formulated.

The typical ICG system is a combination of hardware and software. The hardware includes a central processing unit, one or more workstations (including the graphics display terminals), and peripheral devices such as printers. Plotters, and drafting equipment. Some of this hardware is shown in Figure. The software consists of the computer programs needed to implement graphics processing on the system. The software would also typically include additional specialized application programs to accomplish the particular engineering functions required by the user company.

It is important to note the fact that the ICG system is one component of a computer-aided design system. As illustrated in Figure, the other major component is the human designer. Interactive computer graphics is a tool used by the designer to solve a design problem. In effect, the ICG system magnifies the powers of the designer. This has been referred to as the synergistic effect. The designer performs the portion of the design process that is most suitable to human intellectual skills (conceptualization, independent thinking); the computer performs the task best suited to its capabilities (speed of calculations, visual display, storage of large 81WWts of data), and the resulting system exceeds the sum of its components.

There are several fundamental reasons for implementing a computer-aided design system.

1. To increase the productivity of the designer. This is accomplished by helping the designer to the product and its component subassemblies and parts; and by reducing the time required in synthesizing, analyzing, and documenting the design. This productivity improvement translates not only into lower design cost but also into shorter project completion times.
2. To improve the quality of design. A CAD system permits a more thorough engineering analysis and a larger number of design alternatives can be investigated. Design errors are also reduced through the greater accuracy provided by the system. These factors lead to a better design.
3. To improve communications. Use of a CAD system provides better